



AL2002 Artificial Intelligence BS (CS) Spring 2026

Lab Project

Intelligent Reading Comprehension and Quiz Generation System using Machine Learning

Project Evaluation Rubric Sheet

Evaluation Date: _____
Evaluation Instructor: _____

Reg. No. #: _____	Student Name: _____	Section: _____
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Component	Marks	Key Evaluation Criteria	Self-Evaluation Marks	Marks Awarded
1. Data Analysis & Preprocessing [10 Marks]				
Exploratory Data Analysis	3	Data overview, handle missing value analysis, statistical analysis, outliers detection		
Visualizations	3	Data distribution analysis, correlation analysis, feature relationship		
Preprocessing Pipeline	4	Lowercasing, punctuation removal; Encoding categorical data: One-Hot Encoding features saved or other technique, Feature scaling, Data Cleaning, Feature selection, Feature Engineering, Handling Imbalance data (If present), Data Transformation, Train-Test splits		
2. Model A — Traditional ML [20 Marks]				
≥ 2 Classifiers Implemented	5	Implementation of at least two traditional machine learning classifiers such as Logistic Regression, SVM, Naive Bayes, Random Forest, or XGBoost		
Feature Engineering	5	Appropriate feature engineering techniques applied, such as One-Hot Encoding, TF-IDF, Bag of Words, cosine similarity, or other relevant techniques depending on the dataset		
Question Generation & Answer Verification	5	Meaningful question generation, correct answer prediction/verification. Given (article, question, option) triple, model predicts whether selected option is correct.		
Metric Comparison Table	5	Performance metrics reported and		



		compared across all models on validation/test set using appropriate evaluation metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, or BLEU/ROUGE/METEOR for NLP generation tasks		
3. Model A — Unsupervised / Semi-Supervised [10 Marks]				
Approach Implemented	5	Implementation and evaluation of at least one unsupervised or semi-supervised approach such as K-Means, Label Propagation, or GMM; comparison against supervised models		
Evaluation Metrics	5	Clustering purity, silhouette score, or semi-supervised F1 reported		
4. Model A — Ensemble [05 Marks]				
Ensemble Strategy	3	Soft voting (average probability outputs from SVM + LR + NB), Hard Voting (majority vote across ≥ 3 classifiers), or Stacking (meta-classifier trained on base-model outputs). Code is clean and strategy is clearly named.		
Improvement Demonstrated	2	Ensemble outperforms (or matches with justification) individual models		
5. Model B — Distractor Generation [20 Marks]				
Candidate Extraction Pipeline	4	Phrases extracted from passage via string matching or frequency-based selection		
Feature Engineering for Ranking	4	For each candidate: (a) One-Hot Encoding cosine similarity to correct answer computed, (b) character-level match score computed, (c) passage frequency recorded. TF-IDF cosine similarity used optionally and noted.		
ML Ranker for Distractors	5	Logistic Regression or Random Forest trained to score each candidate; top-3 non-answer candidates selected as distractors; diversity penalty applied so distractors are not trivially similar; model persisted via joblib.		
Plausibility & Diversity	4	Each generated quiz presents three distractors per question; diverse, grammatically consistent, factually wrong. Example (a) appear plausible to an uninformed reader, (b) are definitively wrong w.r.t. the passage, (c) are lexically diverse, and (d) share the same syntactic form as the correct answer.		
Evaluation (BLEU/ROUGE/METEOR)	3	BLEU, ROUGE, METEOR scores reported for distractor generation quality		
6. Model B — Hint Generation [10 Marks]				
Extractive Hint Scorer	5	Each sentence in the passage scored by relevance to the question using; (a) cosine similarity of One-Hot Encoded / sentence-		



		embedding representations, OR (b) Logistic Regression trained on keyword overlap, sentence position, and sentence-length features. Top-K sentences surfaced as ranked hints.		
Graduated Hints (3 Levels)	5	Hint 1 = general, Hint 2 = specific, Hint 3 = near-explicit (sentence containing or closely paraphrasing the answer). All three levels present and correctly ordered in the UI.		

7. User Interface [15 Marks]

Screen 1 — Article Input	3	Text area for pasting or uploading a reading passage; option to load a random RACE dataset sample for quick testing; 'Submit' button triggers both Model A and Model B inference simultaneously; loading indicator shown during inference.		
Screen 2 — Quiz View	4	Generated question displayed on screen; 4 options (A–D); Check button; colour-coded correct/incorrect (proper and functional).		
Screen 3 — Hint Panel	4	Collapsible or tabbed panel with three graduated hints from Model B; hints revealed progressively; 'Reveal Answer' button appears only after all hints have been viewed; UI prevents skipping hints.		
Screen 4 — Analytics Dashboard	2	Model metrics displayed; inference latency shown; CSV export available		
UX & Error Handling	2	All four screens usable without reading a manual; friendly error messages for empty input and model failure; loading indicators during inference; sufficient colour contrast (WCAG AA) and readable font sizes; keyboard navigation possible.		

8. Final Report [05 Marks]

Structure & Completeness	3	All 11 required sections present: Abstract (≤ 200 words), Introduction & Motivation, Related Work (≥ 5 cited papers including RACE, BERT, BLEU/ROUGE originals), Dataset Analysis, Model A Design/Training/Results, Model B Design/Training/Results, UI Description, Evaluation & Discussion, Limitations & Future Work, Conclusion, References. Paper-style formatting.		
Clarity & Discussion	2	Methodology clear; results discussed; limitations and work addressed		

9. Code Quality [05 Marks]

Project Submission	5	Correct Submission of project folder.		
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TOTAL 100 / 100

Key Suggestions (Applied to This Evaluation)

National University of Computer and Emerging Sciences

Fast School of Computing

Project 2026

Islamabad Campus



- **Checkpoints Required:** Groups using Google Colab or other platforms implement model checkpoints.
- **Evaluation Metrics Update:** Not necessary to use Accuracy or Precision as primary metrics for generation tasks. Use **BLEU Score, ROUGE Score, and METEOR Score** for Model A and Model B evaluation.
- Before coming for the demo, kindly fill out this sheet. Also, **make sure to arrive at the demo venue 5 minutes before your scheduled time**. After arriving, directly download your submission from GCR and keep it ready in a working state for the demo.

Evaluator Signature: _____	Final Marks Awarded: _____ / 100
Remarks:	